

Pixels Know When to Go Deep: Lightweight Depth Routing for Autonomous Driving Detectors

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Abstract

001 *Adaptive-depth detectors such as AnyDepth-DETR expose a*
002 *full-depth supernet and a reduced-depth basenet from a sin-*
003 *gle set of weights, but the choice is typically fixed globally*
004 *rather than adapted to scene content. We study per-image*
005 *depth routing: selecting the operating point image by im-*
006 *age to obtain the best accuracy–efficiency tradeoff at any*
007 *target compute budget. A condition-stratified analysis on*
008 *BDD100K shows that the supernet–basenet accuracy gap*
009 *varies up to $10\times$ across scene conditions, with scene spar-*
010 *sity emerging as the single strongest difficulty signal. We*
011 *compare three routing strategies over the binary supernet–*
012 *basenet decision: a condition oracle, detection-confidence*
013 *routing, and pixel-statistics routing using luminance and*
014 *edge density from raw pixels with no GPU involvement.*
015 *On BDD100K, both image-based routers match or exceed*
016 *the condition oracle, with pixel-statistics routing match-*
017 *ing detection-confidence routing at negligible cost. On*
018 *nulImages, pixel-statistics routing outperforms detection-*
019 *confidence routing by capturing scene sparsity via edge*
020 *density where confidence scores are uninformative. These*
021 *results suggest that lightweight pixel-level signals can ef-*
022 *fectively drive per-image depth routing without auxiliary*
023 *models or scene labels. However, the metric-based diffi-*
024 *culty axis does not always coincide with human notions of*
025 *driving difficulty, pointing to safety-driven routing as an im-*
026 *portant open problem.*

027 1. Introduction

028 Perception systems for autonomous driving must operate
029 in real time across a wide range of conditions, from clear
030 daylight to nighttime rain and from sparse motorways to
031 dense urban intersections, while sharing compute with other
032 safety-critical modules. The conventional response is to de-
033 ploy a fixed model at a fixed compute budget, accepting a
034 single accuracy–efficiency tradeoff regardless of what the
035 camera sees. Yet detection difficulty is far from uniform:

the accuracy gap between a deep network and a shallow one
can vary by an order of magnitude depending on lighting,
weather, and crowd density. Choosing the network depth
per image rather than globally would concentrate compute
where it matters most. Figure 1 makes this concrete. In a
hard scene (from the model’s perspective, not the human’s),
the deep network detects 35 objects versus 25 for the shal-
low network, with tighter bounding boxes. In an easy scene,
both networks agree exactly. Scene difficulty determines
whether deeper inference is worth invoking.

We study this question in the context of AnyDepth-
DETR [9], a detector whose backbone, encoder, and de-
coder layers can be selectively skipped at inference time.
From one set of weights it exposes a *supernet* (high accu-
racy, high compute) and a *basenet* (lower compute, mod-
est accuracy loss), separated on BDD100K by an AP gap
of 0.016 at roughly a $1.5\times$ inference speedup. Treating
AnyDepth-DETR as a fixed platform, we ask: *given that*
multiple operating points already exist, which one should
be used for a given image, and what signal should drive that
decision? A condition-stratified analysis on BDD100K [25]
shows that the supernet–basenet accuracy gap varies nearly
 $10\times$ across scene conditions, from 0.0034 to 0.033 AP. To
test whether routing signals generalize, we repeat the analy-
sis on nulImages [1], a predominantly daytime multi-camera
dataset with no scene condition labels.

We compare three routing strategies drawing on differ-
ent signal sources: scene metadata, model output, and raw
pixels. A **condition oracle** on ground-truth scene attributes
(time of day, weather, road type) serves as a condition-level
reference. **Detection-confidence routing** uses the basenet’s
mean top- K detection score as a per-image difficulty proxy,
requiring no labels. **Pixel-statistics routing** derives the sig-
nal entirely from per-image luminance and edge density on
CPU, with no GPU involvement. In continuous-perception
deployment both image-based signals add negligible over-
head: pixel-statistics routing costs only a CPU Sobel filter,
while detection-confidence routing reuses the basenet’s own
output, which the system already computes for every frame.

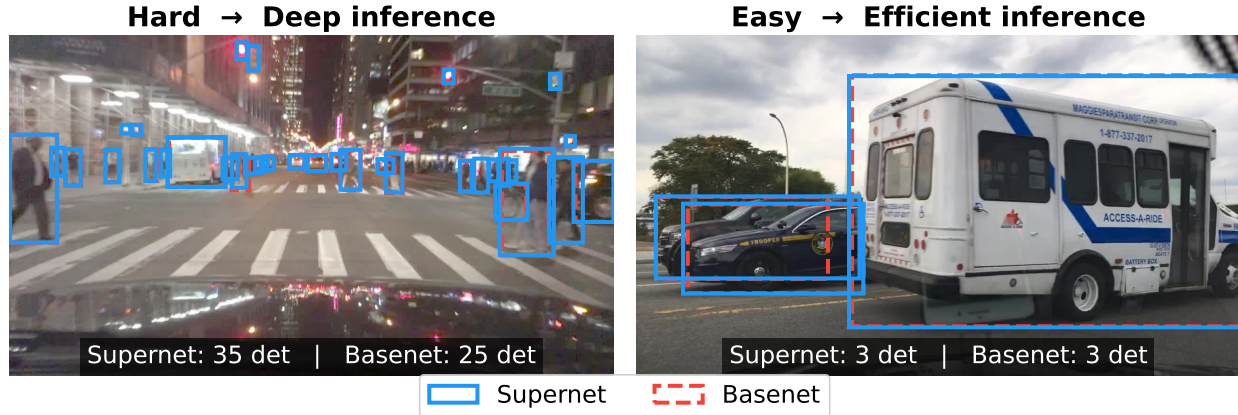


Figure 1. **Adaptive inference depth on BDD100K (AnyDepth-DETR).** **Left:** A scene that is hard for the detector (not necessarily for a human observer), where the supernet (blue, solid) detects 35 objects versus 25 for the basenet (red, dashed), with tighter bounding boxes. **Right:** A scene that is easy for both networks, where they agree on 3 detections and routing to the efficient basenet incurs no accuracy penalty. See Appendix A for more examples.

On BDD100K, both image-based routers match or exceed the condition oracle at matched compute. The gain comes from granularity: per-image signals identify hard frames within easy conditions and easy frames within hard ones, whereas a condition label assigns all matching images to the same network. Pixel-statistics routing matches detection-confidence routing at negligible cost, suggesting that scene complexity is largely recoverable from low-level image content alone. On nuImages, confidence variation is driven by category difficulty across 24 classes rather than scene-level conditions, reducing detection-confidence routing to marginal benefit; pixel-statistics routing recovers the signal by capturing scene sparsity via edge density and outperforms detection-confidence routing. The condition-stratified analysis also identifies crowd sparsity as the single strongest difficulty signal, and edge density is a direct pixel-level proxy for it; this is why the same signal dominates on both datasets despite their very different illumination profiles.

These results suggest that lightweight pixel-level signals can effectively drive per-image depth routing, providing a practical and label-free solution to adaptive inference under the real-time and resource constraints of autonomous driving.

Contributions.

- A condition-stratified analysis on BDD100K revealing up to $10\times$ variation in the supernet–basenet accuracy gap across scene conditions, with scene sparsity emerging as the single strongest difficulty signal.
- A unified routing framework over the binary supernet–basenet decision that compares three strategies (condition oracle, detection-confidence routing, pixel-statistics

routing) as full compute–accuracy curves on two driving datasets: one with scene condition labels and one without.

- The finding that per-image routing beats the condition oracle at matched compute, with edge density as the strongest pixel-level signal on both datasets, matching detection-confidence routing at negligible cost on BDD100K and outperforming it on nuImages.
- The observation that metric-based routing difficulty (sparse, low-clutter scenes) does not always coincide with human notions of driving difficulty, pointing to safety-critical routing objectives as an important open problem.

2. Related Work

Real-time object detection. The YOLO lineage [19, 21] pairs multi-scale feature pyramids [12, 15] with NMS post-processing. End-to-end transformer detectors, starting with DETR [3], eliminate NMS via bipartite matching; DN-DETR [11] and DINO [28] stabilize training through query denoising, and RT-DETR [29] reaches real-time latency. AnyDepth-DETR [9] extends RT-DETR with selectable inference depth from a single set of weights. We treat it as a fixed platform and ask which signal should select among its operating points.

Adaptive-depth inference. Networks that adjust their computation per input span early-exit classifiers [7, 18], spatially adaptive computation [4] and adaptive inference graphs [20], and slimmable [26, 27] and Once-for-All [2] networks supporting multiple widths, depths, and resolutions from a single run (see [5] for a survey). Applied to detection, early exits discard deeper stages and break multi-scale coverage. AnyDepth-DETR [9] addresses this by partitioning each stage into an essential path and a skip-

pable refinement path, preserving the feature hierarchy at every depth. Every operating point is thus a real detector, which makes the binary routing reduction in Section 3.1 well-posed. These works address how to *train* a depth-adaptive network; we study how to *route* between its operating points.

Input-conditioned routing. Several works train an auxiliary network that selects routes per input. SkipNet [22] and BlockDrop [23] learn block-skip policies via reinforcement learning; DynamicDet [13] routes between detector branches with a trained gate; AnytimeYOLO [10] analyzes early-exit configurations under anytime constraints; and AdaDet [24] measures per-image detection uncertainty via stochastic localization and entropy, a model-internal signal that, like detection confidence, depends on detection-score informativeness. All require an auxiliary policy tied to a specific backbone. Our setting shares DynamicDet’s binary-branch structure but replaces the trained router with a scalar threshold over a no-learning signal: either the basenet’s own detection confidence, which it already produces, or per-image luminance and edge density computed on CPU. Their effectiveness rests on the correlation between the routing signal and per-image detection difficulty, a property that low-level statistics preserve across both datasets we study.

Driving perception and temporal coherence. Driving perception runs continuously under real-time budgets shared with other safety-critical modules. Prior work [8] couples adaptive-depth networks with QoS feedback control; VALO [17] provides an anytime framework for LiDAR detection under bounded compute. A complementary line exploits temporal coherence between consecutive frames, propagating features from sparse key frames [30] or amortizing recurrent inference across the sequence [14]. These works adapt *when* to invoke a model; we address *which depth* to invoke per image. The two directions are compatible: because scene conditions persist over many consecutive frames, our routing decisions need not be recomputed every frame and combine naturally with key-frame schemes.

3. Method

3.1. AnyDepth-DETR: A Brief Review

AnyDepth-DETR [9] trains a single detection model that executes at multiple computational depths without retraining, by decomposing the network into *shared* layers and *skippable* layers.

Architecture. The model has three depth-configurable submodules. The **backbone** (ResNet-50) partitions each

Table 1. **Accuracy–efficiency Pareto frontier** across depth configurations on BDD100K val (AP@[0.5:0.95]). BB: backbone (full/skip). Enc: encoder (full/FPN-skip/PAN-skip/skip). Dec: decoder depth. Pareto-efficient configs marked *. FPS measured at batch size 1 on an NVIDIA RTX 6000 Blackwell GPU.

Config	BB	Enc	Dec	AP	FPS
supernet*	full	full	6	0.3751	286
super-d5*	full	full	5	0.3737	294
super-d4*	full	full	4	0.3713	303
enc-fpn-skip	full	FPN	6	0.3684	294
enc-pan-skip	full	PAN	6	0.3680	286
base-enc-full*	skip	full	6	0.3616	345
base-d5*	skip	skip	5	0.3614	400
base-fpn-skip	skip	FPN	6	0.3600	370
base-pan-skip	skip	PAN	6	0.3573	345
basenet*	skip	skip	4	0.3594	417

stage into a fixed block followed by skippable blocks, with a binary skip flag per stage (4 flags total). The **encoder** (HybridEncoder) carries two independent skip flags, one for the FPN top-down path and intra-scale transformer, and one for the PAN bottom-up path. The **decoder** (RTDETRTransformer) executes only the first d of its six cross-attention layers, with $d \in \{4, 5, 6\}$. The full depth space is therefore

$$\Delta = \{0, 1\}^4 \times \{0, 1\}^2 \times \{4, 5, 6\}, \quad (1)$$

yielding $16 \times 4 \times 3 = 192$ configurations from a single set of weights. Further architecture details are provided in Table 10 (Appendix F).

Training. Each batch iteration performs two forward passes under a single optimization step: a *supernet pass* (all skip flags off, $d = 6$) supervised against ground-truth detections with varifocal loss, L1, and GIoU; and a *basenet pass* (all skip flags on, $d = 4$) supervised against the supernet’s outputs via knowledge distillation [6] losses on logits, boxes, and intermediate features. Only the two extremes are trained; intermediate configurations generalize because the shared-weight structure and distillation loss align features across depths without per-configuration retraining. We denote the full-depth supernet and reduced-depth basenet configurations as δ_{sup} and δ_{bas} , respectively, and write $f(x; \delta)$ for the detector output on image x at depth δ , where varying δ trades accuracy for efficiency without retraining.

Accuracy profile and Pareto frontier. AnyDepth-DETR builds on RT-DETR [29], a real-time DETR variant that replaces NMS with bipartite matching. The depth-configurable skip paths above are the only structural addition; the detection head, loss formulation, and matching

procedure are unchanged. On BDD100K val, the supernet reaches AP=0.3751 and the basenet AP=0.3594, a gap of 0.016 AP at roughly a $1.5\times$ inference speedup.

Although $|\Delta| = 192$, the practically useful subset is small. Table 1 profiles ten representative configurations spanning all three depth axes. All four encoder-skip variants fall strictly off the Pareto frontier: bypassing the FPN or PAN path saves only ≈ 0.1 ms while losing 0.005–0.007 AP@[0.5:0.95]. The four encoder-skip configs that fall between the two clusters (AP 0.3573–0.3684) are Pareto-dominated; intermediate Pareto-efficient points exist but cluster near the two extremes, so the practical routing choice reduces to a binary decision between δ_{sup} and δ_{bas} . Both endpoints are practical operating points, as the basenet is a fast model rather than a degraded one, which is what makes per-image routing meaningful.

3.2. Adaptive Depth Routing

Problem statement. Given that the depth space reduces to a binary choice between δ_{sup} and δ_{bas} (Section 3.1), and a compute budget B expressed as the expected fraction of supernet invocations, we seek a routing function

$$R : \mathcal{X} \rightarrow \{\delta_{\text{bas}}, \delta_{\text{sup}}\} \quad (2)$$

that maximizes mean AP subject to

$$\mathbb{E}_{x \sim \mathcal{X}}[\mathbf{1}[R(x) = \delta_{\text{sup}}]] \leq B. \quad (3)$$

The routing function takes the form

$$R(x) = \begin{cases} \delta_{\text{sup}} & \text{if } \phi(x) < \tau, \\ \delta_{\text{bas}} & \text{otherwise,} \end{cases} \quad (4)$$

where $\phi : \mathcal{X} \rightarrow \mathbb{R}$ is a per-image routing signal (low values indicate frames that benefit more from the supernet) and τ is a threshold that sweeps the compute–accuracy tradeoff. Extending this formulation to a richer set of operating points remains future work.

Routing granularity. Condition-level routing assigns all images sharing a scene attribute (e.g., all night images) to the same network, constraining ϕ to take as many distinct values as there are condition labels. Per-image signals, derived from the model’s own output or from raw pixel statistics, operate at full granularity and can identify hard frames within an otherwise easy condition (a sparse nighttime motorway) or easy frames within a hard one (a well-lit rain scene with large, well-separated vehicles). We study both regimes and compare their tradeoffs in Section 4.5.

3.3. Routing Strategies

3.3.1. Condition Oracle

The oracle uses ground-truth BDD100K scene attributes to establish a reference for condition-level routing. From the

condition-stratified analysis (Section 4.2), the two conditions with the largest supernet–basenet gap are nighttime and sparse (crowd-low) scenes. The combined oracle routes accordingly:

$$R_{\text{ora}}(x) = \delta_{\text{sup}} \quad \text{if } x \in \mathcal{C}_{\text{night}} \cup \mathcal{C}_{\text{low}}, \quad (5)$$

where $\mathcal{C}_{\text{night}}$ and $\mathcal{C}_{\text{low}}$ are the night and crowd-low image sets; all other images route to δ_{bas} . Although the oracle relies on ground-truth labels unavailable at inference time, some attributes it uses are readily obtainable via clock and GPS in deployed vehicles; it is not the optimal condition-level allocation (a greedy oracle ranking all conditions by supernet–basenet gap would be tighter), but it captures the two dominant difficulty signals and serves as a strong practically motivated baseline for the image-based routers below.

3.3.2. Detection-Confidence Routing

Detection-confidence routing uses detection scores from the network as a routing signal. For each image, the detector produces a set of object detections, each associated with a classification confidence score; we aggregate these into a single per-image signal as the mean top- K detection score

$$\phi_{\text{conf}}(x) = \frac{1}{K} \sum_{k=1}^K s_k(x), \quad (6)$$

where $s_1(x) \geq \dots \geq s_K(x)$ are the K highest scores. We set $K=20$ to match the approximate mean detection count per image on BDD100K, with results stable across $K \in \{10, 30\}$ in preliminary experiments. A low score indicates detector uncertainty, either because objects are faint (low light, adverse weather) or because the scene is sparse and precision-critical, and triggers escalation to the supernet via Eq. (4). The threshold τ is calibrated on the training set by sweeping the empirical percentiles of ϕ_{conf} . Latency measurements and deployment overhead are reported in Section 4.7.

3.3.3. Pixel-Statistics Routing

Pixel-statistics routing derives the signal from cheap per-image pixel statistics on CPU, incurring $\mathcal{O}(HW)$ time with no GPU involvement. We compute two statistics:

$$\ell(x) = \frac{1}{HW} \sum_{i,j} Y_{ij}(x), \quad (7)$$

$$e(x) = \frac{1}{HW} \sum_{i,j} \sqrt{(G_x)_{ij}^2 + (G_y)_{ij}^2}, \quad (8)$$

where Y is the luma channel in YUV space and G_x, G_y are Sobel gradient responses [16]. $\ell(x)$ proxies time-of-day (low $\rightarrow$ night) and $e(x)$ proxies scene complexity (low $\rightarrow$ sparse or low-contrast). The routing rule escalates to the supernet whenever either signal falls below its threshold:

$$R_{\text{pix}}(x) = \delta_{\text{sup}} \quad \text{if } \ell(x) < \tau_L \text{ or } e(x) < \tau_E, \quad (9)$$

otherwise it routes to δ_{bas} . We evaluate three variants: luminance alone (with τ_E disabled), edge density alone (with τ_L disabled), and the combined router with both active. Each variant is calibrated on the training set by sweeping its active threshold(s) over the empirical percentiles of the corresponding signal; in the combined router, τ_L and τ_E are stepped together at the same percentile of their respective distributions.

4. Experiments

4.1. Setup

Datasets. We conduct primary experiments on BDD100K [25], a large-scale autonomous driving dataset with 70K training and 10K validation images at 1280×720 resolution and 10 object categories. Each image carries scene-level attributes (time of day, weather, scene type) that enable condition-stratified analysis. To assess generality beyond a single domain, we additionally evaluate on nuImages [1], a multi-camera driving dataset with 60K training and 14K validation images at 1600×900 resolution, drawn from all six camera views, covering 24 object categories. Unlike BDD100K, nuImages lacks scene condition labels and is captured predominantly in daytime, providing a complementary testbed under uniform illumination. All results report AP@[0.5:0.95] in COCO format.

Training and Evaluation. For BDD100K we fine-tune from a COCO-pretrained checkpoint at 800×480 input resolution, using a $3 \times$ schedule (36 epochs), and AdamW ($\text{lr} = 10^{-4}$, batch size 16). For nuImages we apply the same recipe at 1024×576 input resolution. For routing experiments we run both networks on the full validation set once and evaluate router-mixed prediction sets offline, allowing threshold sweeping at negligible cost. Thresholds are fixed by training-set calibration; the reported supernet invocation rates are observed on val. Latency is measured as mean per-image GPU inference time on a single NVIDIA RTX Pro 6000 Blackwell GPU (batch size 1, 20 warm-up batches). Qualitative examples, per-class AP, full threshold sweep tables, and architecture details are provided in the supplementary appendix.

4.2. Condition-Stratified Analysis

Table 2 reports supernet and basenet AP across 12 condition groups on BDD100K val, with the gap quantifying the per-condition benefit of deeper inference.

Two dominant patterns emerge. First, adverse lighting amplifies the depth benefit: nighttime shows a gap of $+0.0262$, three times the daytime gap of $+0.0087$. In low light the basenet’s shallower feature hierarchy fails to aggregate sufficient context for reliable localization, and the

supernet’s additional layers recover this lost precision. Second, scene sparsity is the single strongest routing signal: crowd-low images yield a gap of $+0.0330$, nearly ten times the crowd-high gap of $+0.0034$. In sparse scenes objects tend to be small and distant, so a single missed detection or imprecise bounding box incurs a large fractional AP penalty; the supernet’s deeper feature hierarchy improves both recall and localization of these difficult objects, yielding a disproportionately large absolute gain. Highway ($+0.0193$) and clear-weather ($+0.0191$) scenes follow the same pattern of open, low-clutter environments where localization precision is the binding constraint.

Across all 12 condition groups the supernet–basenet AP gap spans a tenfold range, from 0.0034 to 0.0330, confirming that no single global operating point fits every condition. Notably, what is hard for the detector does not always match human-perceived difficulty: a sparse daytime motorway looks easy to a human yet yields one of the largest gaps in Table 2. Nighttime and crowd-low are the two strongest condition-level signals and motivate the combined oracle in Section 4.3. Scene sparsity also leaves a direct footprint in raw pixels: low crowd density is measurable as edge density without semantic labels, which we exploit in Section 4.4.

4.3. Condition Oracle Routing

Using the combined condition oracle from Eq. (5), images in the night or crowd-low groups are routed to the supernet and all others to the basenet. As shown in Table 3, this routes 67% of images to the supernet and achieves $\text{AP}=0.3704$, recovering $+0.0110$ AP over always-basenet while remaining -0.0047 AP below always-supernet. The residual gap reflects the oracle’s coarse granularity: entire condition groups share one routing decision, so easy images within a hard condition (and hard images within an easy one) are systematically misrouted. This gap sets the target for the image-based routers below.

4.4. Image-based Routing Strategies

Figure 2 shows the AP–compute tradeoff as the threshold τ sweeps the empirical percentiles of each signal on both datasets, with the corresponding oracles marked for reference. Full per-step threshold values and AP at each operating point are listed in Table 8 and Table 9 in Appendix D.

Detection-confidence routing. On BDD100K, basenet detection scores range from 0.184 to 0.873 (median 0.653), with low-confidence images concentrated at night and in sparse scenes, naturally aligning with the oracle’s routing targets. At $\tau = 0.706$ (66.7% supernet) the router achieves $\text{AP}=0.3717$, exceeding the combined oracle ($\text{AP}=0.3704$) through per-image granularity: a sparse daytime motorway, for instance, triggers escalation because its detection score is low, even though its condition label (daytime,

Table 2. **Condition-stratified AP on BDD100K val** (10 K images). The supernet–basenet gap quantifies the per-condition benefit of deeper inference, varying by nearly $10\times$ across groups (from $+0.0034$ in crowd-high to $+0.0330$ in crowd-low). Scene sparsity is the strongest difficulty signal, larger than lighting and weather effects. The crowd split uses the per-image object count median as threshold.

Condition	N	Supernet	Basenet	Gap
All	10,000	0.3751	0.3594	+0.0156
<i>Time of day</i>				
Daytime	5,258	0.3884	0.3797	+0.0087
Night	3,929	0.3491	0.3229	+0.0262
Dawn/Dusk	778	0.3580	0.3451	+0.0129
<i>Weather</i>				
Clear	5,346	0.3520	0.3329	+0.0191
Overcast	1,239	0.4118	0.3996	+0.0122
Rainy	738	0.3581	0.3444	+0.0137
Snowy	769	0.3590	0.3447	+0.0143
<i>Scene type</i>				
Highway	2,499	0.3431	0.3238	+0.0193
City street	6,112	0.3607	0.3470	+0.0137
Residential	1,253	0.4122	0.4000	+0.0122
<i>Crowd density</i>				
Low ($\leq$ median)	5,025	0.4041	0.3711	+0.0330
High ($>$ median)	4,975	0.3605	0.3571	+0.0034

clear) would route it to the basenet under the oracle. At $\tau = 0.785$ (90.5% supernet), $AP = 0.3767$ surpasses always-supernet (0.3751), since per-image detector performance is non-monotonic in depth and the router exploits both optima simultaneously.

Pixel-statistics routing. As Figure 2(a) shows, the pixel-statistics curve tracks the detection-confidence curve closely on BDD100K across the full compute range. Edge density alone at 66.6% supernet achieves $AP = 0.3716$ ($+0.0012$ vs. oracle), matching detection-confidence routing to within 0.0001 AP at zero GPU cost. The combined OR rule at 69.4% supernet yields $AP = 0.3711$ ($+0.0007$ vs. oracle). Luminance alone is a slightly weaker proxy but remains competitive ($+0.0002$ vs. oracle at 62% supernet): low luminance correlates with nighttime, which carries a $+0.0262$ supernet benefit (Table 2), and it is this signal–difficulty correlation that makes mean luminance an effective routing cue. Edge density captures the complementary and stronger signal: low edge density is a direct pixel-level proxy for low crowd density, which is the strongest routing signal in Table 2 (gap $+0.0330$ vs. $+0.0034$ for crowd-high). Sweeping τ_E thus implicitly ranks images by scene sparsity, routing the frames where the supernet gains most first, which explains edge density’s slight advantage over luminance on BDD100K.

4.5. Routing Strategy Comparison

Table 3 consolidates all strategies on BDD100K at a comparable compute budget ($\approx 67\text{--}70\%$ supernet invocations). Both image-based routers match or exceed the GT-label oracle: bootstrap 95% CIs on the AP difference (500 resamples of val image IDs) are $[+0.0000, +0.0016]$ for detection-confidence and $[+0.0005, +0.0025]$ for edge density, confirming that per-image granularity identifies hard frames more precisely than a discrete condition label shared across thousands of images. Pixel-statistics routing achieves this using only lightweight CPU computation, indicating that scene complexity is sufficiently encoded in low-level image statistics.

4.6. Generalization to nuImages

To evaluate robustness to domain shift, we repeat the routing analysis on nuImages using a model trained with the identical training recipe used for BDD100K. Lacking scene-condition labels, we substitute crowd density (per-image object count vs. per-dataset median) as the oracle signal. This oracle routes 54.9% of images to the supernet and achieves $AP = 0.2707$ (vs. 0.2597 for always-basenet and 0.2964 for always-supernet).

Dataset statistics. Figure 3 shows the distributional difference between the two domains. BDD100K’s bimodal luminance distribution reflects its balanced day/night split;

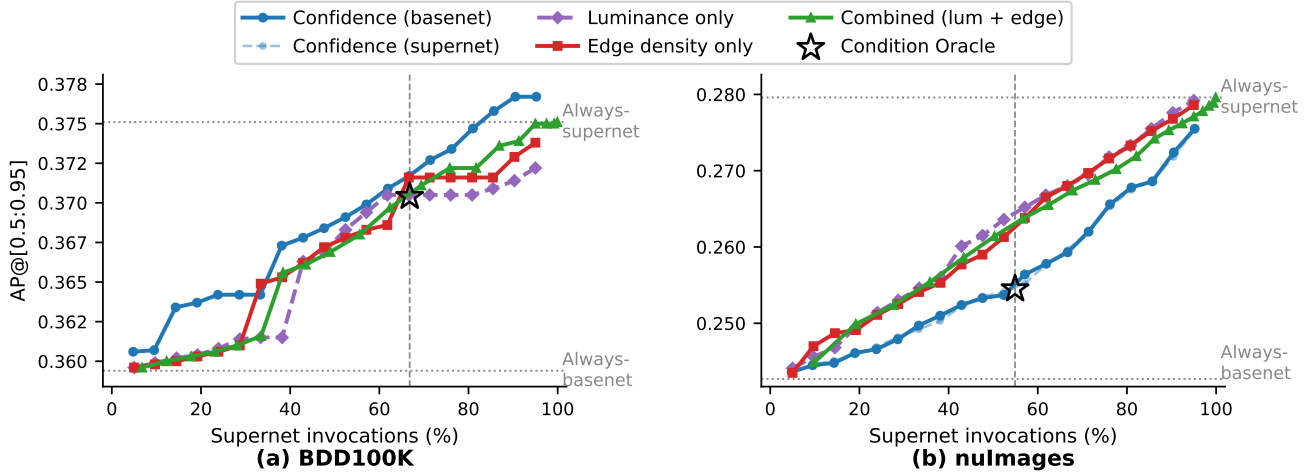


Figure 2. **AP vs. supernet invocation rate** on BDD100K (a) and nuImages (b). Sweeping the routing threshold τ traces a continuous compute-accuracy curve for each strategy. Dotted lines mark always-supernet and always-basenet AP; the dashed vertical line marks the oracle compute budget. On BDD100K, detection-confidence routing is the dominant signal: basenet scores spread widely across night and sparse scenes, correlate strongly with per-image supernet benefit, and match or exceed the condition oracle (GT labels; 67% supernet, AP=0.370) at the same budget. On nuImages, detection-confidence routing provides only marginal benefit because confidence variation is driven by category difficulty across 24 classes rather than scene-level conditions; pixel-statistics routing recovers the signal through scene sparsity and outperforms detection-confidence routing by up to +0.009 AP. Per-step threshold values and AP at each operating point are listed in Appendix D.

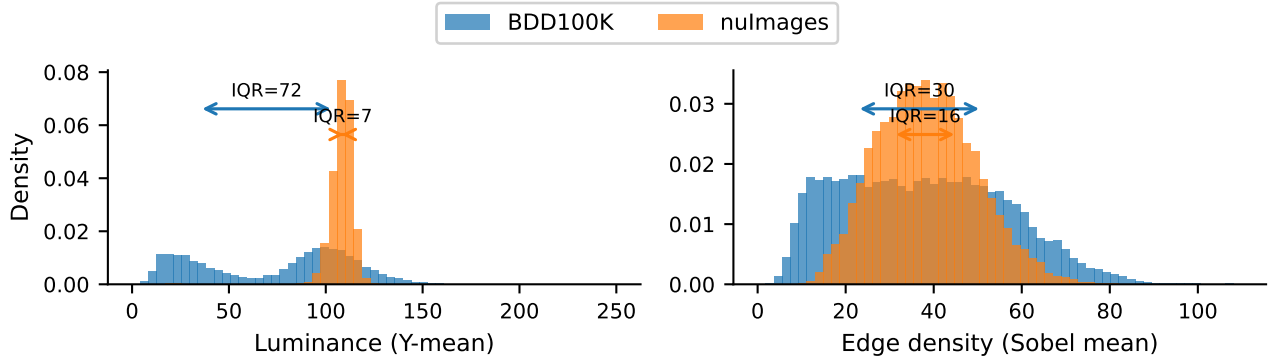


Figure 3. **Pixel-statistics distributions** for BDD100K (blue) and nuImages (orange) validation sets, computed on resized inputs. BDD100K’s bimodal luminance distribution (IQR=72) reflects its balanced day/night split; nuImages shows a narrower daytime-concentrated peak (IQR=7). Edge density is more widely spread across both datasets (IQR=30 vs. 16) than luminance. However, distributional width does not determine routing quality: a signal is effective when it correlates with per-image detection difficulty regardless of its absolute range, as confirmed by both edge density and luminance being effective routing signals on nuImages, with edge density the stronger of the two (Table 4).

454 nuImages shows a narrower daytime-concentrated peak.
 455 This difference in distributional width does not predict routing
 456 quality. What matters is whether a signal *correlates with*
 457 *per-image detection difficulty*, i.e. the degree to which the
 458 supernet outperforms the basenet on a given image. A signal
 459 with a narrow range can still be highly discriminative if
 460 its variation aligns with variation in difficulty.

Routing results. Table 4 reports all strategies at $\approx 52\%$
 supernet invocations, close to the 54.9% oracle budget.
 Detection-confidence routing provides only marginal ben-
 efit on nuImages: the basenet variant barely exceeds the
 oracle (+0.0006) and the supernet variant falls slightly be-
 low it (−0.0003). Unlike BDD100K, where low confidence
 concentrates in nighttime and sparse scenes that directly
 benefit from deeper inference, on nuImages confidence is

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Table 3. **Routing strategy comparison on BDD100K val.** %S: percentage of supernet invocations. ΔO and ΔS : signed AP differences vs. the condition oracle and always-supernet. *Cost*: information used for the routing decision.

Router	Cost	%S	AP	ΔO	ΔS
Sup.-only	—	100	0.3751	+0.0047	—
Base.-only	—	0	0.3594	−0.0110	−0.0157
Oracle	GT labels	67	0.3704	—	−0.0047
Det.-confidence	Base. fwd	67	0.3717	+0.0013	−0.0034
Luminance	Y-mean	62	0.3705	+0.0002	−0.0045
Edge density	Sobel mag.	67	0.3716	+0.0012	−0.0035
Combined (LVE)	Pixels	69	0.3711	+0.0007	−0.0040

Table 4. **Routing strategy comparison on nuImages val** (1024×576 input). Lacking scene-condition labels, the oracle uses crowd density (supernet if object count $\leq$ per-dataset median; 55% supernet invocations). All image-based routers are evaluated at $\approx 52\%$ supernet invocations, close to the oracle budget. %S: percentage of supernet invocations. ΔO : AP difference vs. the crowd oracle. ΔS : AP difference vs. always-supernet. *Cost*: information used for the routing decision.

Router	Cost	%S	AP	ΔO	ΔS
Sup.-only	—	100	0.2964	+0.0257	—
Base.-only	—	0	0.2597	−0.0110	−0.0367
Oracle	Obj. count	55	0.2707	—	−0.0257
Det.-conf. (base)	Base. fwd	52	0.2712	+0.0006	−0.0252
Det.-conf. (super)	Sup. fwd	52	0.2704	−0.0003	−0.0260
Luminance	Y-mean	52	0.2779	+0.0072	−0.0185
Edge density	Sobel mag.	52	0.2795	+0.0088	−0.0169
Combined (LVE)	Pixels	50	0.2778	+0.0071	−0.0186

driven by category difficulty across 24 object classes rather than scene-level conditions: supernet–basenet gaps range from +0.063 for rigid buses down to −0.003 for ambulances (full per-class breakdown in Appendix C), so a single routing threshold cannot simultaneously optimize for both. As shown in Appendix E, the nuImages confidence distribution is centered lower and more spread than BDD100K’s (IQR 0.28 vs. 0.19), yet this variation does not align with per-image supernet benefit. All three pixel-statistics routers exceed the oracle by a wide margin: edge density +0.0088, luminance +0.0072, combined +0.0071. Edge density is the strongest router on nuImages because it captures scene sparsity as a pixel-level proxy for crowd density, the same condition-level signal that dominates on BDD100K (Table 2), confirming that this correlation transfers across both datasets regardless of illumination.

4.7. Routing Overhead

Table 5. **Per-image latency** on BDD100K (800×480) and nuImages (1024×576). [†]No additional forward pass required: the detection task runs continuously on every frame, and confidence scores are obtained as a byproduct.

Method	nuImages	BDD100K
Supernet inference (GPU)	11.6 ms	9.8 ms
Basenet inference (GPU)	8.0 ms	7.1 ms
Pixel-stats routing (CPU)	2.2 ms	
Det.-confidence routing	<0.1 ms [†]	

Table 5 reports per-image inference latency. Pixel-statistics routing adds a fixed 2.2 ms CPU overhead per frame (a Sobel filter on the already-resized input tensor) with no GPU involvement. Detection-confidence routing adds negligible overhead: confidence scores are a byproduct of the basenet pass already performed for continuous per-frame detection. Furthermore, since scene conditions change slowly, the routing decision can be reused across consecutive frames, amortising even this cost.

5. Conclusion and Future Work

Per-image depth routing works because the accuracy gap between the supernet and basenet varies with scene content, and any signal that correlates with this per-image difficulty can effectively drive routing decisions. On BDD100K, detection-confidence routing and pixel-statistics routing both exceed the condition oracle, with edge density matching detection-confidence routing at negligible cost. On nuImages, where confidence variation is driven by category difficulty rather than scene-level conditions, edge density recovers the difficulty axis through scene sparsity and outperforms detection-confidence routing by a clear margin. These results suggest that lightweight pixel-level signals can effectively drive per-image depth routing, offering a practical path to adaptive inference under the real-time and resource constraints of autonomous driving.

A notable finding is that the metric-based routing difficulty (sparse, low-clutter scenes) does not exactly coincide with human notions of driving difficulty, opening a compelling research question: how should routing objectives be extended to incorporate safety-critical priorities alongside AP-based gain? Coupling routing decisions with scene-level risk is an important direction for future work. As adaptive-depth object detectors become more widely available, evaluating whether pixel-level routing signals transfer across architectures and domains will be a natural next step.

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